

Meridian Health and Wellness Network

Job Task Analysis & Cognitive Task Analysis

Registered Nurse

Client:

Meridian Health and Wellness Network (Fictional)

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Conducted by

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1.1 Role Overview

Job Title	Registered Nurse — Medication Administration
Department	Clinical Nursing Division, Meridian Health and Wellness Network
Reporting to	Charge Nurse / Unit Nurse Manager
Shift Pattern	12-hour rotating shifts (day/night). Peak medication rounds at 07:00, 12:00, 18:00, 22:00.
Patient Load	Average 4–6 patients per nurse per shift. Up to 8 during peak census periods.
Analysis Method	Semi-structured SME interview (two experienced RNs with 8+ years tenure), direct observation of two medication rounds, and review of incident reports and EHR system logs.

1.2 Task Inventory with DIF Ratings

#	Task	Knowledge, Skills & Attitudes Required (KSAs)	D	I	F	Priority
1	Receive and verify medication order in EHR	EHR navigation; clinical order interpretation; awareness of patient allergy flags	3	3	3	HIGH
2	Cross-reference patient allergy and medication interaction records	Pharmacology knowledge; critical thinking; EHR allergy module navigation	3	3	3	HIGH
3	Prepare medication (draw, measure, label)	Aseptic technique; dosage calculation; label verification protocol	3	3	3	HIGH
4	Verify 5 Rights at patient bedside (Right patient, drug, dose, route, time)	Patient identification protocol; barcode scanning; verbal confirmation skills	2	3	3	HIGH
5	Administer medication (oral, IV, topical, etc.)	Clinical administration technique; route-specific protocols; infection control	3	3	3	HIGH
6	Document administration in EHR immediately post-administration	EHR documentation module; accurate timestamping; clinical notation standards	2	3	3	HIGH
7	Monitor patient for adverse reactions (15-minute post-administration check)	Clinical observation skills; adverse reaction recognition; escalation protocols	3	3	2	HIGH
8	Respond to and document medication error or near-miss event	Error reporting protocol; non-punitive reporting culture awareness; EHR incident module	3	3	2	HIGH
9	Communicate medication administration to oncoming shift during handoff	SBAR communication framework; verbal and EHR documentation consistency	2	2	3	MODERATE
10	Manage medication reconciliation for admitted or transferred patients	Reconciliation protocol; collaboration with pharmacy; patient history review	3	3	1	MODERATE

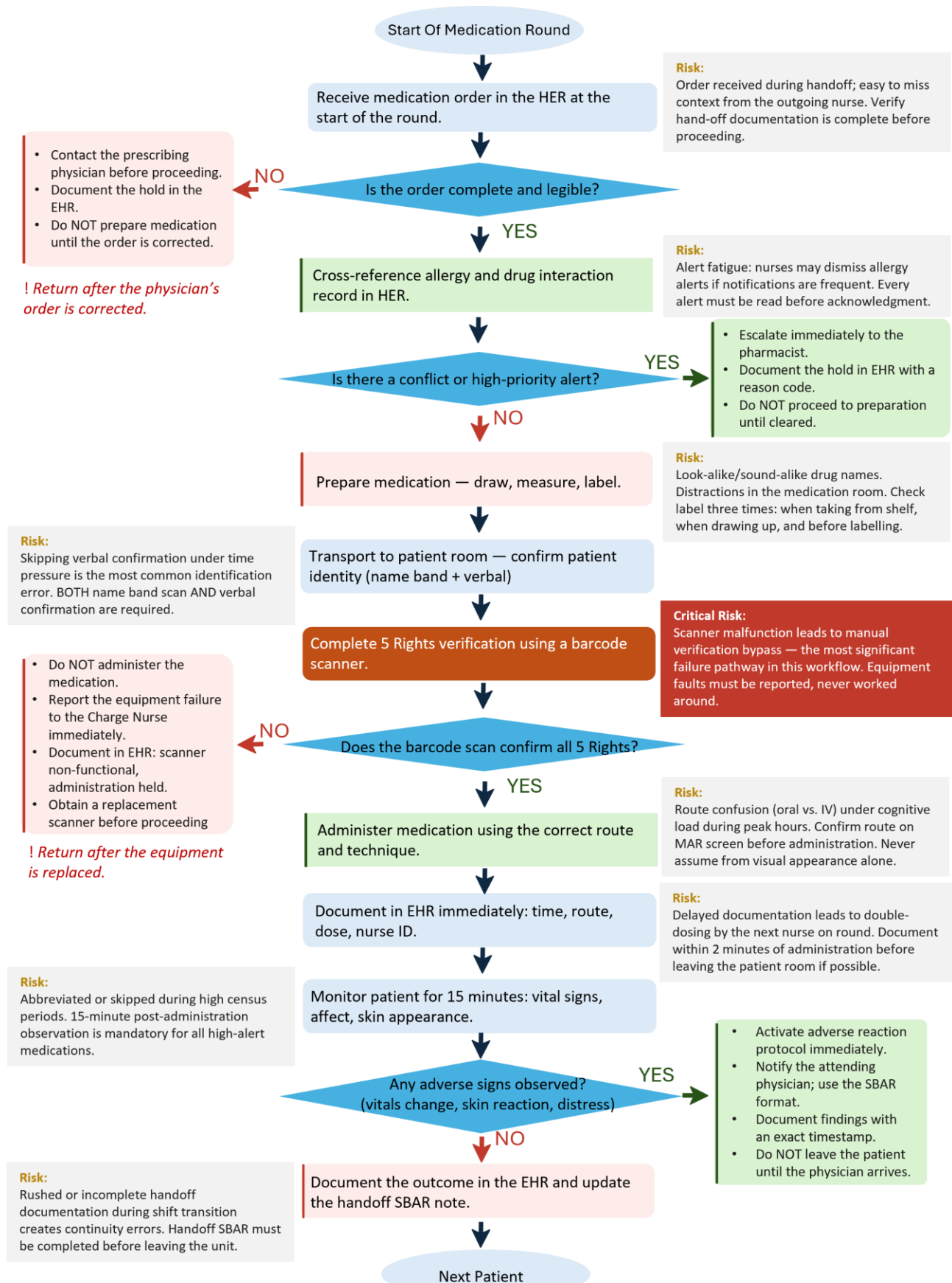
1.3 Cognitive Task Analysis (CTA)

The CTA applied specifically to Tasks 1, 2, 4, 7, and 8, which involve the highest degree of clinical judgment and decision-making. SME interviews were used to surface tacit knowledge that would otherwise remain invisible in job descriptions or standard SOPs.

Cognitive Demand	What the Nurse Must Recognize	Decision Point	Risk if Missed
Cue Recognition	Unusual dosage in EHR order; unfamiliar drug name; conflicting allergy alert	Hold administration and contact prescribing physician vs. proceed with caution	Medication error, patient harm, liability exposure
Alert Prioritization	Distinguishing actionable alerts from routine/low-priority EHR notifications	Escalate alert to pharmacist vs. acknowledge and proceed	Alert fatigue leading to missed critical warnings
Time Pressure Management	Multiple patients with concurrent medication windows; competing urgent tasks	Prioritize by clinical urgency vs. scheduled time vs. patient acuity	Delayed or missed medication window; protocol deviation
Adverse Reaction Detection	Subtle changes in patient affect, skin, breathing, or vital post-administration	Classify as expected side effect vs. adverse reaction requiring intervention	Delayed treatment of adverse reaction; patient harm
Error Recovery	Realizing a verification step was skipped, or a wrong route was used	Self-report immediately vs. delay reporting vs. not report	Harm escalation; non-reporting creates system-level blind spots

1.4 Process Map: Medication Administration Workflow

The process map below represents the end-to-end medication administration workflow as observed and confirmed by SMEs. Decision points represent moments where nurse judgment determines the path forward. Critical error risk points are highlighted.



1.5 Performance Gaps Identified Through JTA/CTA

Gap	Root	Recommended Intervention
EHR 5 Rights verification step frequently abbreviated under time pressure	Training + System	Scenario-based simulation training for high-pressure rounds + EHR hard-stop alert for skipped verification
Alert fatigue causing nurses to dismiss critical EHR notifications	System	Non-training: EHR alert tiering review with IT + pharmacy (reduce non-actionable alerts)
Post-administration documentation delayed or incomplete	Training + Process	Job aid at point of care + EHR documentation workflow redesign
Inconsistent adverse reaction monitoring (15-minute check)	Training	Refresher training with clinical scenarios + structured observation checklist
Under-reporting of errors due to fear of disciplinary action	Organizational	Non-training: Leadership communication + anonymous reporting mechanism